

UNIVERSITY OF CALIFORNIA

Los Angeles

Expanding the characterization of hot Jupiter atmospheres using high-resolution
spectroscopy

A dissertation submitted in partial satisfaction
of the requirements for the degree
Doctor of Philosophy in Astronomy & Astrophysics

by

Luke Michael Finnerty

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PREVIEW

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ABSTRACT OF THE DISSERTATION

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Professor Michael P. Fitzgerald, Chair

As exoplanet science moves from discovery to detailed characterization of individual planets, ongoing advances in ground-based instrumentation are enabling precise constraints on atmospheric properties of extrasolar planets for the first time. In particular, ultra-stable high-resolution infrared spectroscopy can directly probe the emission spectra of close-in giant exoplanets via high-resolution cross-correlation spectroscopy (HRCCS). High-resolution observations resolve individual molecular lines, enabling constraints on molecular abundances, isotopologue ratios, wind speeds, and atmospheric circulation patterns. By directly detecting the planetary atmosphere in emission, the HRCCS technique enables novel characterization of non-transiting planets, and is less impacted by the presence of clouds or hazes than transmission-based techniques.

HRCCS requires stable, well-calibrated instruments. The Keck Planet Imager and Characterizer (KPIC) provides this via a single-mode fiber feed from the Keck II AO system to the Keck/NIRSPEC high-resolution spectrograph. KPIC features a large number of moving parts and necessitates a complicated daytime calibration procedure. I developed a set of

software scripts and procedures for this task, enabling reliable daytime calibration of KPIC within 90 minutes. This reliability has made KPIC a workhorse for characterizing the atmospheres of both hot Jupiters and directly-imaged substellar companions in the H , K , and L bands. I used KPIC observations of the ultra-hot Jupiter WASP-33 b to demonstrate the feasibility of HRCCS analysis with KPIC, confirming the presence of a thermal inversion and measuring abundances of CO and H₂O using an atmospheric retrieval pipeline I developed. Building on this success, I began a large survey of hot Jupiter atmospheres, with a goal of obtaining a large homogeneous data set suitable for constraining the underlying compositional diversity of the hot Jupiter population.

Results from this survey so far include the first bounded measurements of the C/O ratio and metallicity for the benchmark hot Jupiter HD 189733 b, free retrieval of the H₂O vertical mixing profile for the ultra-hot Jupiter KELT-20 b, the first atmospheric detection of the non-transiting hot Jupiter HD 143105 b, joint K and L -band characterization of the benchmark hot Jupiter HD 209458 b, a tentative detection of the warm Neptune GJ 436 b, and a comparative analysis of six ultra-hot Jupiters. I also place limits on the atmospheric properties of the eccentric super-Jupiter HD 80606 b using seeing-limited observations from Keck/NIRSPEC.

In addition, the KPIC hot Jupiter survey has detected at up to 16 additional hot Jupiters in K band thermal emission, and analysis of these targets is ongoing. Continued observational efforts aim to extend the characterization of several of these targets to the L band, and to broaden to the observed phase coverage of several ultra-hot Jupiters in order to constrain global circulation patterns. These results have already significantly expand our knowledge of hot Jupiter atmospheres and will continue to bear scientific fruit in the future, particularly by providing a benchmark set of atmospheric compositions to compare to planet formation models and begin unraveling the origins of hot Jupiters.

The dissertation of Luke Michael Finnerty is approved.

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University of California, Los Angeles

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PREVIEW

For my grandmother, Dr. Victoria Finnerty, who got me started

And for Peyton, who convinces me to consider the flowers

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